Panduan Dashboard — POWER BI (Detail, klik-demi-klik)

Tugas Besar Big Data · Credit-Card Default (Taiwan 2005) · jalur ELT (`bigdata_elt`)

> Power BI membaca warehouse **ELT** (`bigdata_elt`) via Hive ODBC. Dashboard memakai 4 tabel `kpi_*`. Acuan desain: `DASHBOARD_SPEC.md`. Preview hasil: `docs/DASHBOARD_PREVIEW.pdf`.
> (Versi Tableau ada di `docs/PANDUAN_DASHBOARD_DETAIL.pdf`.)

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1. Prasyarat — siapkan server dulu

1. Stack Hive hidup, dan **matikan layanan yang tidak perlu** agar HiveServer2 lega (kunci kestabilan):
   `powershell docker compose stop spark-master spark-worker kafka kafka-ui docker compose up -d --force-recreate hiveserver2 # sesi bersih docker compose ps # cukup: postgres + hive-metastore + hiveserver2`
2. **Hive ODBC driver 64-bit** (Cloudera/Microsoft) terpasang.
3. DSN ODBC sudah dikonfigurasi: Host `localhost`, Port `10000`, Database **`bigdata_elt`**, Hive Server Type **HiveServer2**, Mechanism **User Name** user `hive`, **Thrift Transport = Binary**, **SSL = OFF** → **Test** → **SUCCESS**. (Detail: `dashboard/HIVE_ODBC_CONNECTION.md`.)

2. Tabel sumber (4 KPI)

Tabel	Kolom
`kpi_overall`	`default_rate`, `total_clients`
`kpi_default_by_demographic`	`dimension`, `category`, `clients`, `default_rate`
`kpi_monthly_default_vs_macro`	`date_key`, `month_name`, `default_rate`, `exchange_rate_twd_usd`, `real_broad_eer`, `total_reserves`
`kpi_corr_default_macro`	`corr_fx`, `corr_reer`, `corr_reserves`

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3. Menghubungkan data — pilih SATU cara

Cara A — ODBC Import (paling stabil, disarankan)

1. **Home** → **Get Data** → **More…** → **ODBC** → **Connect**.
2. Pilih **DSN** Anda → **OK**.
3. Saat ditanya **Import / DirectQuery** → **pilih `Import`**.
4. Di **Navigator**, buka database `bigdata_elt`, **centang hanya 4 tabel `kpi_*`** → **Load**.
   (Jangan tarik `elt_fact_monthly`/`elt_client_trends` 900k baris.)

Cara B — ODBC DirectQuery (jika diminta query live)

1. Get Data → ODBC → DSN → **DirectQuery** → centang **hanya `kpi_*`** → Load.
2. **WAJIB setel agar handle tidak gugur** (`Invalid OperationHandle`):
   - **File** → **Options and settings** → **Options** → **DirectQuery** → **"Maximum connections per data source" = 1** (query berurutan, tidak berebut).
   - **Options** → **Data Load** → hilangkan centang **"Allow data previews to download in the background"**.
   - DSN → Configure → Advanced: **Rows Fetched Per Block = 10000**, **Use Native Query** ON.
3. Jangan jalankan Spark/Kafka selama dashboard aktif (lihat §1).

Cara C — CSV (fallback tanpa ODBC, sudah disiapkan)

File ada di `dashboard/exports/`:
1. **Get Data** → **Text/CSV** → pilih `bigdata_elt__kpi_overall.csv`, `bigdata_elt__kpi_default_by_demographic.csv`, `bigdata_elt__kpi_monthly_default_vs_macro.csv`, `bigdata_elt__kpi_corr_default_macro.csv` → **Load** (ulangi per file).

> Regenerasi CSV: `python dashboard/export_kpis_csv.py` (atau lihat perintah beeline di guide).

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4. Siapkan model & tipe data

1. **Model view** (ikon kiri) → cek tipe tiap kolom:
   `default_rate` / `corr_*` / kolom makro = **Decimal number**; `total_clients/clients/date_key` = **Whole number**; `dimension/category/month_name` = **Text**.
2. **Urutkan bulan Apr→Sep:** klik tabel `kpi_monthly_default_vs_macro` → pilih kolom `month_name` → **Column tools** → **Sort by column** → **`date_key`**.
3. **Hindari penjumlahan salah:** untuk `default_rate/corr_*`, di pane **Data** klik kolomnya → **Column tools** → **Summarization = Average** (atau **Don't summarize**), bukan Sum.

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5. Membangun 4 zona (visual per zona)

Zona A — Kartu KPI (visual Card)

1. Pane **Visualizations** → **Card**. Tarik **`kpi_overall[default_rate]`** ke **Fields**.
2. **Format (ikon kuas)** → **Callout value** → **Value decimal places = 2**. Agar tampil persen: di pane Data pilih `default_rate` → **Column tools** → **Format = Percentage, 2 desimal** (hasil **22,12%**).
3. Card kedua: **`kpi_overall[total_clients]`** → Format ribuan → **150.000**.
4. Card ketiga–kelima: **`corr_fx`**, **`corr_reer`**, **`corr_reserves`** (2 desimal → **0,00**).
5. Tiap Card: **Format** → **General** → **Title** beri judul ("Overall Default Rate", dst).

Zona B — Default rate per demografi (Clustered bar chart + slicer)

1. Visual **Clustered bar chart**. **Y-axis = `kpi_default_by_demographic[category]`**, **X-axis = `default_rate`** (set **Average**).
2. Tambah **Slicer** terpisah: field **`dimension`** → Format slicer → **Single select** ON. Memilih `sex/education/marriage/age_band` akan memfilter bar.
3. **Garis referensi overall:** pilih bar chart → **Analytics pane** → **Average line / Constant line** = `0.2212` → beri label "overall".
4. **Data labels** ON. Urut: "…" (More options) → **Sort axis** → **default_rate** → **Descending**.

Zona C — Tren bulanan Apr–Sep 2005 (Line chart, sumbu ganda)

1. Visual **Line chart**. **X-axis = `kpi_monthly_default_vs_macro[month_name]`** (sudah sort by `date_key`).
2. **Y-axis = `default_rate`** (Average).
3. **Secondary y-axis:** tarik **`exchange_rate_twd_usd`** ke **Secondary y-axis** (atau pakai *Line and clustered column chart* → kolom = makro, garis = default_rate).
4. **Format** → **Y-Axis** → **Secondary** aktif; beri warna garis berbeda (merah=default, biru=kurs).
5. (Pilih makro) buat **Field parameter**: **Modeling** → **New parameter** → **Fields** berisi `exchange_rate_twd_usd`, `real_broad_eer`, `total_reserves` → jadikan **Slicer** pemilih indikator.

Zona D — Default vs makro (Scatter chart, korelasi ≈ 0)

1. Visual **Scatter chart**. **X Axis = `exchange_rate_twd_usd`** (Average),
   Y Axis = `default_rate` (Average), **Values = `month_name`** (agar 1 titik per bulan).
2. **Analytics pane** → **Trend line** ON.
   Tambah **Card** kecil `corr_fx` di dekatnya sebagai anotasi korelasi.

> Titik akan mendatar → **korelasi** ≈ **0,00** (temuan jujur: label default per-klien konstan tiap bulan — keterbatasan grain, report §10).

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6. Filter interaktif (wajib §8)

Tambah **Slicer** di kanvas (otomatis memfilter semua visual sehalaman):
- **`dimension`** (Single select) → zona B.
- **Indikator makro** (Field parameter) → zona C & D.
- **`month_name`** (rentang/checklist) → zona C & D.
- Atur cakupan via **Format** → **Edit interactions** bila perlu membatasi visual yang terpengaruh.

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7. Menyusun & merapikan halaman

1. Susun: 4–5 **Card** sejajar di atas; **bar** (kiri) & **line** (kanan) di tengah; **scatter** + **slicer** di bawah. Atur posisi/ukuran dengan drag.
2. Tambah **Text box** judul: "Credit-Card Default Risk Taiwan 2005 — ELT (Power BI)".
3. **View** → **Themes** untuk warna konsisten (samakan dengan versi Tableau agar screenshot sebanding).

8. Simpan & ekspor (untuk report §8–§9)

- **File** → **Save As** → `dashboard/powerbi_elt_dashboard.pbix`.
- Screenshot: **File** → **Export** → **PDF**, atau **Win+Shift+S** → simpan `dashboard/screenshots/powerbi_elt_dashboard.png`.
- Verifikasi angka: **overall** ≈ **0,2212**, **total clients 150.000** — harus sama dengan Tableau (ETL). Kecocokan = bukti konsistensi ETL vs ELT (report §9).

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9. Troubleshooting (kasus nyata di proyek ini)

Gejala	Sebab	Solusi
`Invalid OperationHandle … EXECUTE_STATEMENT` / "Failed to save modifications"	DirectQuery: query paralel berebut handle / sesi HS2 di-recycle	**Max connections per data source = 1**; matikan background preview; muat **hanya `kpi_*`**; **Refresh** setelah HS2 di-recreate; atau pakai **Import** / CSV
"Unexpected response received from server"	DSN transport salah	DSN: **Thrift Transport = Binary**, **SSL = OFF**, Mechanism **User Name** `hive`
Query lambat / HS2 berat	Spark/Kafka ikut jalan / tarik tabel 900k	`docker compose stop spark-master spark-worker kafka kafka-ui`; jangan DirectQuery tabel fakta
Nilai ke-jumlah (mis. default_rate aneh besar)	agregasi Sum	set **Summarization = Average**
Bulan tak urut	belum sort	`month_name` **Sort by column** `date_key`

Tetap gagal ODBC	driver/timeout	**Cara C — CSV** di `dashboard/exports/` (pasti jalan)

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10. Ringkas urutan

1. `docker compose stop spark-master spark-worker kafka kafka-ui` + recreate HS2 →
2. Get Data → ODBC (**Import**, DB `bigdata_elt`) → centang **kpi_** → Load →
3. set tipe & Average, sort bulan →
4. bangun zona A–D →
5. tambah slicer →
6. rapikan + Save `.pbix` + screenshot →
7. cek overall 0,2212 == Tableau.